

STEI-ZStat: A Low-Cost AD5941 Potentiostat for Electrochemical Impedance Spectroscopy

Krishudson Clement
Institut Teknologi Bandung
Bandung, Indonesia
clement.teja@outlook.com

Kent Frumentius
Institut Teknologi Bandung
Bandung, Indonesia
kent.frumentius@gmail.com

Iqbal Fawwaz Ramadhan
National Yang Ming Chiao Tung University
Hsinchu, Taiwan
iqbalfawwazr.st12@nycu.edu.tw

Muhammad Ogih Hasanuddin
Inha University
Incheon, South Korea
Institut Teknologi Bandung
Bandung, Indonesia
moginh@inha.edu

Wervyan Shalannanda
Institut Teknologi Bandung
Bandung, Indonesia
Johannes Kepler University Linz
Linz, Austria
K12239569@students.jku.at

Isa Anshori
Biomedical Engineering Study Program
School of Elect. Eng. & Informatics
Institut Teknologi Bandung
Bandung, Indonesia
isaa@staff.stei.itb.ac.id

Abstract—Electrochemical impedance spectroscopy (EIS) enables label-free biosensing, but benchtop instrumentation remains a barrier. Published validations of low-cost alternatives often rely on spectral overlays rather than predefined acceptance criteria. This work presents STEI-ZStat, a single-channel AD5941 potentiostat (0.2 Hz–200 kHz) on a 63.1×45.0 mm board, costing US\$25.36 ($n=5$). We validated it against a PalmSens4 reference using a Randles dummy cell via a four-stage procedure (reactance-sign screening, resistive plateau checks, CNLS recovery, and equivalence testing). Seven replicates were acquired per instrument. Both reproduced nominal resistance within 0.31%, with comparable median replicate $|Z|$ standard deviations (0.024% vs. 0.026%). STEI-ZStat showed lower median magnitude deviation (0.111% vs. 0.236%). CNLS fitting recovered parameters within $\pm 2\%$ resistance and $\pm 10\%$ capacitance limits, with all cross-instrument 95% confidence intervals within margins. Inter-board reproducibility was high (RSDs: 0.027%, 0.011%, 0.148% for R_s , R_{ct} , C_{dl}). The principal limitation was high-frequency phase error (1.21° at 200 kHz vs. 0.33° for PalmSens4). Validation was limited to passive dummy cells, excluding biological or redox measurements.

Keywords—Electrochemical Impedance Spectroscopy (EIS), Analog Front-End (AFE), Potentiostat, Instrument Validation, Biosensor Instrumentation, AD5941

I. INTRODUCTION

Electrochemical impedance spectroscopy (EIS) is a widely used method for label-free biosensing [1] because it characterizes interfacial properties, such as solution resistance (R_s), charge-transfer resistance (R_{ct}), and double-layer capacitance (C_{dl}), with minimal system perturbation [2]. Unlike cyclic voltammetry, which applies a potential sweep that can drive unwanted faradaic processes, EIS typically applies a small-signal excitation ($5\text{--}15\text{ mV}_p$) around the rest potential. This allows functionalized surfaces to be measured repeatedly without degradation [1], [3]. To capture these dynamics, an instrument must cover a broad frequency range (typically 0.01 Hz to 100 kHz): the upper decades resolve the solution-resistance plateau and the high-frequency side of the interfacial semicircle, while the lower decades capture slow interfacial and diffusional processes [1], [2], [4], [5].

The high cost and large footprint of benchtop wide-band EIS instruments limit their deployment in point-of-care, field,

and educational settings [6]–[8]. While many low-cost, open-source potentiostats have been developed to address this limitation [9]–[11], most are restricted to voltammetric methods. However, modern integrated analog front-ends, particularly the Analog Devices AD5941, now make wide-band EIS feasible on compact, low-power boards [8], [12].

Despite hardware advances, demonstrating quantitative agreement using predefined acceptance criteria remains inconsistent. As summarized in Table I, validation in low-cost EIS systems relies primarily on spectral overlays, error bounds, regression, or hypothesis testing. However, these methods fall short: significance testing ignores practical importance, accuracy bounds do not verify actual compliance, and visual overlays obscure systematic errors. Furthermore, studies rarely apply explicit criteria to flag invalid data. Consequently, we must move beyond mere statistical detectability to establish practical equivalence through a rigorous, predefined validation framework.

To address this gap, this work presents STEI-ZStat, an AD5941-based potentiostat costing US\$25.36 (0.2 Hz–200 kHz) on a 63.1×45.0 mm board. We also introduce a four-stage validation procedure: reactance-sign screening, resistive plateau checks, complex nonlinear least-squares (CNLS) parameter recovery, and equivalence testing against a commercial reference. Performance was evaluated using a precision Randles dummy cell, comparing measurements against a factory-calibrated PalmSens4 reference and analytical spectra predicted from nominal component values. Board-to-board reproducibility was assessed via supplementary measurements on three independently assembled units. Table I compares STEI-ZStat with prior systems; cost-reporting methodologies are detailed in the footnotes. Validation using physical biosensors is deferred to future work.

II. SYSTEM DESIGN

STEI-ZStat is a single-channel, three-electrode potentiostat interface based on the Analog Devices AD5941 analog front-end (AFE) and the Seeed Studio XIAO RP2040 microcontroller unit (MCU) (Fig. 1). The hardware architecture is derived from the design in [12] and optimized to fit within

TABLE I
COMPARISON TABLE OF LOW-COST EIS-CAPABLE POTENTIOSTATS

Name	Freq. Range (Hz)	Reported Cost ^a (USD)	Dimensions (mm)	Electroanalytical Methods	Validation Methods
Pruna et al. [6]	0.1–10,000	300 ^b	120 × 70	EIS	Regression against VMP3, $R^2 = 0.98$
ABE-STAT [13]	0.1–100,000	105 ^c	89 × 74	CV, DPV, OCP, EIS	RMS noise; potentiometric regression
Matsubara [14]	0.1–1,000,000	410 ^{c,f}	Not stated	CV, SWV, CA, CP, EIS	Error bounds only
TBISTAT [7]	1–10,000	77.4 ^d	Not stated	EIS	t-test; Levene; normality checks
ACEStat [15]	1–120,000	60 ^d	68 × 26	CV, LSV, SWV, cSWV, CA, EIS	Randles fitting; margin of error
HELPStat [8]	0.15–200,000	50.5 ^e	61.7 × 44.4	EIS	Paired t-test; Theil–Sen
Hunstat2 [12]	0.2–200,000	30 ^{e,g}	80 × 50 ^h	CV, OCP, EIS	Visual overlay only
STEI-ZStat (TW)	0.2–200,000	25.36 ^e	63.1 × 45.0	EIS ⁱ	Reactance-sign screen; resistive plateau check; CNLS recovery; equivalence test

^a Costs in USD are as reported by each source and have not been renormalized; this work's figure is per unit at a build quantity of five (May 2026). ^b Details not stated. ^c Components only. ^d Components and board fabrication. ^e Components, fabrication, and assembly. ^f Calculations included a \$279 oscilloscope in the BOM. ^g Recalculated from the published bill of materials and stated manufacturing costs; approximately \$27.5. ^h Taken from the open-sourced board design files. ⁱ CV, OCP, CA, SWV, and DPV are also implemented in firmware but are not experimentally validated in this work.

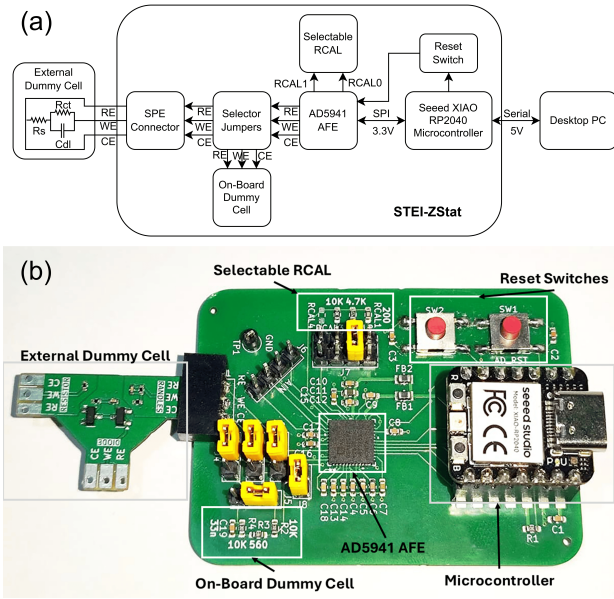


Fig. 1. (a) Block diagram and (b) photograph of STEI-ZStat with Randles dummy cell connected.

a footprint of 63.1×45.0 mm. The AD5941 integrates a potentiostatic control loop with a transimpedance input stage and is the basis for several reported potentiostats [8], [9], [12]. It supports impedance measurements from 0.015 Hz to 200 kHz and excitation voltages of up to $1.2 V_{pp}$ [16], although this study experimentally validates the instrument only from 0.2 Hz to 200 kHz. Compared with the base HunStat2 design, STEI-ZStat introduces three hardware modifications to improve EIS performance.

The first modification is a jumper-selectable R_{CAL} resistor array with values of 200 Ω , 4.7 k Ω , and 10 k Ω , along with an unpopulated footprint for alternative resistance values. During EIS measurements, the known R_{CAL} impedance is measured separately to determine the unknown impedance [17]. The selectable R_{CAL} values allow the calibration resistor to match the expected impedance of the device under test more closely, as recommended by Analog Devices.

The second modification is an on-board dummy cell comprising a Randles circuit and a 10 k Ω (1%) resistor

for self-testing, selectable via jumpers. Although the on-board Randles circuit uses the same nominal component values as the external dummy cell shown in Fig. 1, the external cell was used for validation because it is populated with tighter-tolerance components (R_s : $560 \Omega \pm 0.1\%$, R_{ct} : $10 \text{ k}\Omega \pm 0.02\%$, C_{dl} : $33 \text{ nF} \pm 5\%$), providing a more reliable reference.

The third modification is the PCB layout, which uses the partitioned pinout of the AD5941 to separate analog and digital signal paths. The microcontroller and digital traces are physically isolated from sensitive analog nodes, including electrode connections, calibration resistors, and reference capacitors. This layout maintains a continuous ground plane and is intended to minimize electromagnetic interference (EMI) coupling into high-frequency EIS measurements.

The firmware is implemented in C/C++ using the Arduino IDE and is based on the Analog Devices AD5940/AD5941 library and example code [17], while the Python GUI uses Tkinter and Matplotlib. The software also supports exporting measurement data to a comma-separated values (CSV) file. In addition to EIS, cyclic voltammetry (CV), open-circuit potential (OCP), chronoamperometry (CA), square-wave voltammetry (SWV), and differential pulse voltammetry (DPV) have been implemented but have not yet been experimentally validated.

The unit build cost is US\$25.36 for a build quantity of five (May 2026). This includes PCB fabrication and assembly of the AFE and passive components (US\$20.34), the MCU (US\$4.80), screen-printed electrode (SPE) connectors (US\$0.09), and switches, jumpers, and pin headers (US\$0.13). The total excludes an enclosure, cables, electrodes, a host computer, and assembly labor. Cost and dimensional comparisons against previously reported systems are summarized in Table I.

III. VALIDATION METHODOLOGY

A precision passive circuit minimizes the uncertainty of the test sample, making instrument variability the dominant source of measurement variation [18]. Screen-printed electrodes, in contrast, vary between units and with repeated use [11], [19], so an electrode-based comparison would confound electrode variability with instrument performance.

A dummy cell was therefore used to evaluate STEI-ZStat against a commercial reference before biosensor validation. It was treated as an instrument-metrology fixture rather than a model of a particular biological medium or analyte. A factory-calibrated PalmSens4 (PALM-PS4.F2.10), capable of EIS measurements from 10 μ Hz to 1 MHz, was used as the commercial reference.

The external dummy cell was designed to fit into the SPE connector of both instruments (Fig. 1). It implemented a Randles circuit ($R_s + (R_{ct} \parallel C_{dl})$) with $R_s = 560 \Omega$ ($\pm 0.1\%$), $R_{ct} = 10 \text{ k}\Omega$ ($\pm 0.02\%$), and $C_{dl} = 33 \text{ nF}$ ($\pm 5\%$, YAGEO CC0402JRX7R9BB333). The topology was selected because it is a widely used first-order representation of an electrochemical interface. It also matches the commercial verification cell from which the acceptance criteria were adapted [20] and can be realized entirely from discrete components with specified nominal values. Warburg or constant-phase elements were not included because they would add parameters not directly realized by the discrete components and would complicate the ideal-component validation.

The component values were selected to test distinct regions of the EIS response. They yield high- and low-frequency plateaus of approximately 560 Ω and 10.56 k Ω with a characteristic frequency of approximately 482 Hz. Thus, the 0.2 Hz–200 kHz sweep captured both resistive plateaus and the main semicircular response. The selected nominal values are also consistent with those of a previously reported study of potentiostat performance [21]. The nominal component values were used as the analytical reference. However, the capacitor was not independently measured in the assembled circuit and therefore could not serve as a capacitance reference. The cell was received 134 days before measurement, providing a lower bound on the time since reflow. At 0.2 Hz, the computed impedance magnitude of this circuit is within 0.001% of $R_s + R_{ct}$, so lower frequencies would provide negligible additional information.

EIS measurements used a nominal excitation of 30 mV_p (21.21 mV_{rms}), 0 VDC bias, and 55 logarithmically spaced frequencies swept from 200 kHz to 0.2 Hz. Because of the 12-bit DAC resolution of the AD5941, the actual excitation voltage generated by STEI-ZStat was 30.23 mV_p. Because the dummy cell is passive and linear, this 0.8% excitation difference does not affect impedance. Seven replicate spectra were acquired per instrument.

The instruments used different acquisition strategies. PalmSens4 integrated for the longer of 0.5 s and two excitation periods per frequency, giving a sweep duration of approximately 111 s. STEI-ZStat used frequency-dependent discrete Fourier transform (DFT) integration windows of 0.02–13.67 s and separately measured the load and R_{CAL} , giving a sweep duration of approximately 280 s. The longer low-frequency integration time reduced the effective noise bandwidth of STEI-ZStat relative to PalmSens4, and the shorter high-frequency windows increased it. Integration time and noise bandwidth were therefore not matched, and should be considered when comparing precision.

PalmSens4 used automatic current ranging from 10 mA to 1 nA. STEI-ZStat used the 4.7 k Ω R_{CAL} as the closest available value to the geometric mean of the expected 560–10,560 Ω range. The installed R_{CAL} measured 4696 Ω on

a handheld meter with a specified accuracy of $\pm(0.2\% + 3 \text{ digits})$ at 0.001 k Ω resolution, so the nominal 4.7 k Ω value was retained in firmware. Because the AD5941 measures impedance relative to R_{CAL} , the 1% tolerance of R_{CAL} limits the absolute accuracy of the measured impedance magnitude; higher-accuracy R_{CAL} characterization was not performed. The combined transimpedance amplifier (TIA) gain was automatically adjusted from 500 Ω to 15 k Ω using discrete RTIA and PGA settings.

The validation procedure comprised four stages. Measurement data were first subjected to reactance-sign screening for positive imaginary impedance ($Z'' > 0$), which is inconsistent with the expected capacitive response of the passive Randles cell. Such points were flagged and retained with weight 0.05 rather than removed. Resistive plateaus were evaluated at 200 kHz ($|Z| \approx R_s$) and 0.2 Hz ($|Z| \approx R_s + R_{ct}$). At 200 kHz, residual reactance makes the ideal $|Z|$ approximately 0.10% above R_s ; this offset is included in the reported deviation.

Full-band performance was evaluated by complex nonlinear least-squares (CNLS) regression using the Levenberg–Marquardt algorithm in the lmfit library (v1.3.4) in Python [22]. An ideal capacitor was used rather than a constant-phase

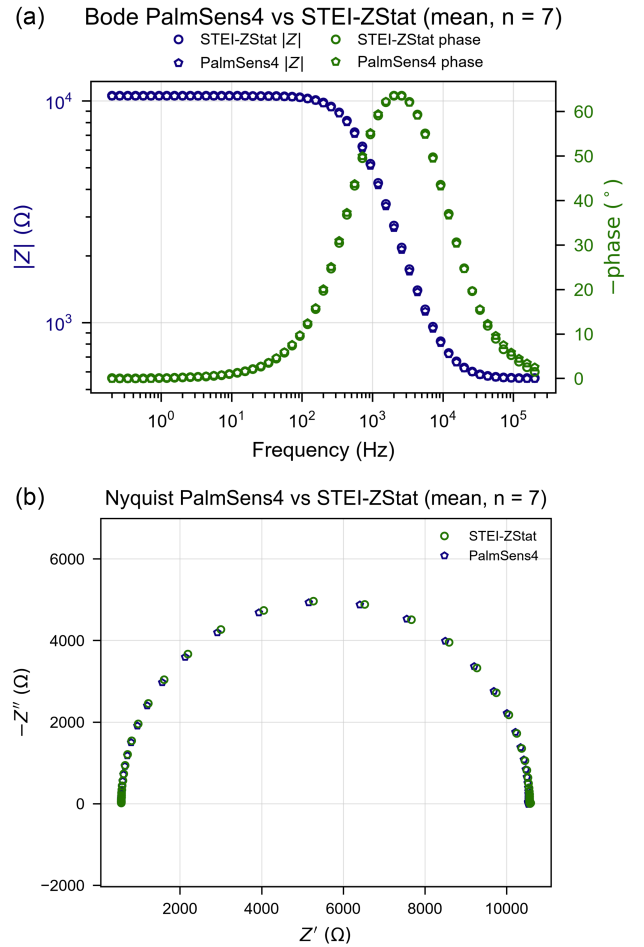


Fig. 2. Overlaid and aggregated seven EIS measurements for STEI-ZStat and PalmSens4 in a (a) Bode plot and a (b) Nyquist plot across 55 frequencies. Error bars are omitted because the replicate standard deviation, with a median of 0.024% in $|Z|$ for STEI-ZStat and 0.026% for PalmSens4, is smaller than the plot markers; replicate spread is reported in Table II.

element so that the fit tested the ideal-component assumption. Real and imaginary components were fitted simultaneously with residuals weighted by $1/|Z|$, with flagged points retained at weight 0.05. Two fits were performed.

The first fit fixed R_s and R_{ct} at their nominal values and fitted C_{dl} to characterize spectral deviation from the nominal Randles model. Because C_{dl} was not independently characterized in circuit, the fitted value was treated as a model parameter rather than a traceable capacitance measurement. The fitted impedance spectrum was compared with the measured spectrum, and the median deviation, maximum deviation, and standard deviation (SD) of $|Z|$ and phase were calculated.

The second fit optimized R_s , R_{ct} , and C_{dl} to assess parameter recovery and cross-instrument equivalence. Parameter deviations were calculated relative to the nominal component values, and replicate variability was summarized using relative standard deviation (%RSD). The resulting values were then compared with predefined acceptance criteria adapted from the Gamry Universal Dummy Cell operator's manual [20]. The manual specifies absolute acceptance windows of 196–204 Ω , 2.95–3.07 k Ω , and 0.90–1.10 μF for the fitted parameters of its own verification cell, with nominal values of 200 Ω , 3.01 k Ω , and 1 μF , respectively. These correspond to $\pm 2\%$ for the resistive elements and $\pm 10\%$ for the capacitance. The same percentages were applied to the present dummy cell. These are vendor tolerances rather than a published standard, are not specifications of the PalmSens4 used in this study, and should not be interpreted as standardized accuracy requirements for EIS instrumentation.

Device equivalence was assessed from the second CNLS fit. Because the seven measurements per instrument were acquired in separate blocks rather than paired trials, the samples were treated as independent ($n = 7$ per instrument). Mean differences and 95% confidence intervals (CI) were calculated using the Welch–Satterthwaite method. Because the population variance is unknown and is estimated from $n = 7$ replicates, critical values were taken from the Student's t distribution rather than the normal distribution, with Welch–Satterthwaite degrees of freedom, using SciPy (v1.18.0). Equivalence was accepted when the entire 95% CI remained within the predefined acceptance window. Although a 90% interval corresponds to two one-sided tests at $\alpha = 0.05$, the 95% CI criterion was intentionally conservative. The resulting limits were $\pm 11.20 \Omega$ for R_s , $\pm 200.00 \Omega$ for R_{ct} , and $\pm 3.30 \text{ nF}$ for C_{dl} , derived from the nominal component values.

Unit-to-unit reproducibility was assessed four weeks after the primary measurements, using three assembled STEI-ZStat boards measured under the same conditions on the same dummy cell, with seven spectra from each board analyzed to match the replicate count of the primary comparison. The microcontroller module was transferred between boards, so this comparison isolates the analog front end, calibration resistor, and board layout.

Because only one unit of each instrument was evaluated in the primary comparison, the equivalence result applies only to the tested STEI-ZStat and PalmSens4 under the specified passive-circuit, excitation, and acquisition conditions. It does not establish unit-to-unit reproducibility or equivalence of the

TABLE II
REPEATABILITY AND PASSIVE-REFERENCE AGREEMENT

Statistics	STEI-ZStat	PalmSens4
200 kHz plateau reading vs. R_s (560.0 Ω)	(561.55 $\pm$ 0.26) Ω , +0.28%	(558.25 $\pm$ 1.02) Ω , –0.31%
0.2 Hz plateau reading vs. $R_s + R_{ct}$ (10560.0 Ω)	(10589.6 $\pm$ 3.2) Ω , +0.28%	(10534.1 $\pm$ 1.0) Ω , –0.25%
Replicate SD, median(max) $ Z $ (%)	0.024 (0.077)	0.026 (0.254)
Replicate SD, phase ($^\circ$); median(max)	0.013 (0.046)	0.025 (0.153)
Impedance deviation against reference, $\Delta Z $ (%); median(max)	0.111 (0.613)	0.236 (0.738)
Phase deviation against reference, Δphase ($^\circ$); median(max)	0.067 (1.215)	0.070 (0.329)

TABLE III
CNLS PARAMETER RECOVERY AND PRACTICAL EQUIVALENCE

Instrument	Parameters	R_s	R_{ct}	C_{dl}
STEI-ZStat	Value	562.35 Ω	9996.03 Ω	30.25 nF
	Deviation	+0.419%	–0.040%	–8.33%
	%RSD	0.010	0.007	0.079
PalmSens4	Value	559.47 Ω	9967.86 Ω	30.94 nF
	Deviation	–0.094%	–0.321%	–6.23%
	%RSD	0.027	0.004	0.263
STEI-ZStat vs. PalmSens4	Difference	+2.875 Ω	+28.17 Ω	–0.693 nF
	95% CI	[+2.73, +3.02]	[+27.47, +28.87]	[–0.77, –0.62]
	Acceptance Criteria	$\pm 11.20 \Omega$ (2% of 560 Ω)	$\pm 200.00 \Omega$ (2% of 10 k Ω)	$\pm 3.30 \text{ nF}$ (10% of 33 nF)

Values are means of seven independent CNLS fits, one per replicate spectrum; %RSD is the relative standard deviation across those seven fits, in percent. Acceptance criteria are fixed from the nominal component values and were set before analysis.

STEI-ZStat design as a whole.

IV. RESULTS

No STEI-ZStat spectrum contained points with $Z'' > 0$. Every PalmSens4 spectrum contained six such points below 1 Hz (0.2–0.72 Hz). These points differed from the analytical reference by only –0.05% to –0.11% in $|Z|$ and approximately 0.05 $^\circ$ in phase, indicating a small sign reversal near the noise level. The same frequencies were affected in all seven replicates, so this behavior was repeatable under the tested PalmSens4 configuration. The points were therefore flagged and down-weighted rather than removed.

Figure 2 shows that the aggregated spectra from both instruments are visually similar. Because the two traces coincide at this scale, the quantitative comparison is made in Fig. 3 and Tables II and III rather than by inspection of the overlay. Both instruments reproduced the nominal resistance values within 0.31% (Table II), well within the predefined $\pm 2\%$ acceptance margin. Replicate variability was also low, with median standard deviations of 0.024% versus 0.026% in $|Z|$ and 0.013 $^\circ$ versus 0.025 $^\circ$ in phase for STEI-ZStat and PalmSens4, respectively. The maximum magnitude SD was 0.077% for STEI-ZStat and 0.254% for PalmSens4, occurring between 1.2 and 1.5 kHz.

Figure 3 shows the deviation from the analytical reference across frequency. STEI-ZStat showed a lower median magnitude deviation than PalmSens4 (0.111% versus 0.236%), while its phase deviation above 10 kHz was larger, as reported below; maximum magnitude deviations were comparable (0.613% versus 0.738%). Median phase deviations were also similar (0.067 $^\circ$ versus 0.070 $^\circ$). Above 10 kHz, STEI-ZStat

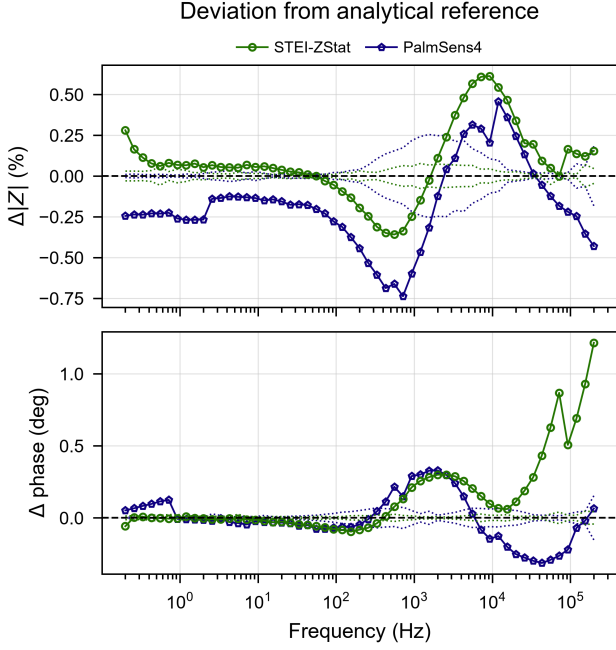


Fig. 3. Deviation of $|Z|$ and phase from the analytical reference plotted against frequency. The analytical reference is the impedance computed from the Randles model with R_s and R_{ct} held at their rated values and C_{dl} fitted; deviation is expressed as a percentage of the reference $|Z|$ and as a phase difference in degrees. Solid traces represent a mean of seven replicate measurements while dotted lines represent ± 1 standard deviation from a single measurement as a noise scale.

phase deviation increased to 1.215° at 200 kHz, compared with 0.329° for PalmSens4. Over the measured points below 100 kHz (0.2 Hz–92.8 kHz), the maximum STEI-ZStat phase deviation was 0.868° at 71.9 kHz.

Mean CNLS relative residuals were 0.36% for STEI-ZStat and 0.33% for PalmSens4, indicating comparable agreement with the fitted model. All fitted parameters satisfied the $\pm 2\%$ resistance and $\pm 10\%$ capacitance criteria, with replicate variability below 0.3% RSD (Table III). Both instruments estimated the capacitance below the 33 nF nominal value (-8.33% and -6.23%). The fitted-parameter differences were statistically significant, as all 95% CIs excluded zero, but every CI remained within the predefined acceptance windows. No confidence interval extended beyond 27% of its corresponding acceptance margin.

Across the three STEI-ZStat boards, mean recovered parameters were 562.43, 562.50, and 562.72 Ω for R_s ; 9995.1, 9997.4, and 9996.1 Ω for R_{ct} ; and 30.32, 30.24, and 30.25 nF for C_{dl} , giving inter-board relative standard deviations of 0.027%, 0.011%, and 0.148%, respectively. The board used in the primary comparison reproduced its previously recovered R_s and R_{ct} to within 0.02%. High-frequency phase deviation was also reproducible between boards, reaching 1.227° , 1.205° , and 1.240° at 200 kHz.

V. DISCUSSION

STEI-ZStat provides EIS performance comparable to PalmSens4 over the validated 0.2 Hz–200 kHz range. At US\$25.36 per unit, it is the least expensive system listed in Table I under the stated cost bases, although the margin over HunStat2 [12] is only about 8% against its recalculated

cost basis, and HunStat2 supports additional voltammetric methods. Its compact footprint and frequency range are also within the range of previously reported systems. Thus, the main contributions of this work are not the hardware cost, size, or frequency range, but the quantitative validation procedure used to assess measurement performance.

Measured repeatability was comparable between the instruments despite their different integration strategies (Table II). The longer STEI-ZStat integration windows at low frequencies may have contributed to this result, so the precision comparison should be interpreted in light of the differences in acquisition settings. Similarly, the lower median analytical magnitude deviation of STEI-ZStat does not indicate uniformly lower measurement error, because its high-frequency phase error was substantially larger.

Both estimates fall outside the component's $\pm 5\%$ tolerance but in the same direction, which is consistent with the aging of the Class II X7R dielectric after reflow. A maximum aging rate of 3% per decade-hour is specified for X7R dielectrics [23], which, over the 134-day interval between receipt of the assembled dummy cell and measurement, corresponds to up to approximately 6% downward drift. However, no aging rate was available for the specific component, so this explanation cannot be verified from the present data. The DC bias was 0 V, so DC-bias effects were absent. The 21 mV_{rms} excitation was also well below the 1 V_{rms} condition at which the capacitance was specified, but the effect of excitation amplitude on the actual capacitance was not characterized here.

Because the capacitor could not be independently measured in the assembled dummy cell, its nominal 33 nF value is the least certain parameter of the analytical reference. The 0.69 nF difference between the instruments' fitted capacitances, 2.3% of the STEI-ZStat value, therefore remains unresolved and is reported rather than attributed to either instrument.

The main STEI-ZStat limitation is high-frequency phase error, which reaches 1.215° at 200 kHz compared with 0.329° for PalmSens4. Such behavior is consistent with the sensitivity of high-frequency EIS to parasitic impedances in the measurement path. The three assembled boards reproduced this deviation within 0.04° at 200 kHz and reproduced the shape of the deviation curve across the upper decade, indicating a systematic error rather than an isolated defect in one board. An error that is systematic in this sense is, in principle, correctable by calibration or compensation, although the responsible mechanism was not isolated here. The 200 kHz boundary should therefore be interpreted cautiously, particularly for measurements whose conclusions depend on small phase differences.

The validation separates statistical significance from practical equivalence. All fitted-parameter differences were statistically significant, but every 95% CI remained within the predefined acceptance window. This distinction is important because significance testing can identify small systematic differences without determining whether they are practically relevant. The present approach quantifies both the difference and its uncertainty, and then determines whether the 95% CI remains within a predefined acceptance window.

The validated range covers much of the frequency region

commonly used for biosensing, including frequencies up to approximately 100 kHz reported in previous applications. However, the present validation begins at 0.2 Hz rather than the 0.01–0.1 Hz reported in some applications; because the AD5941 is specified to 0.015 Hz, this shortfall is a limitation of the validation rather than of the hardware. The validation also used 30 mV_p rather than the 5–15 mV_p excitation typical of small-signal biosensing, so the results do not directly establish performance at lower excitation amplitudes. Only a passive Randles cell with an impedance range of 560 Ω–10.56 kΩ, with a single time constant and 0 V DC bias, was evaluated. The results therefore establish instrument equivalence for the tested passive impedance range, not equivalent performance with functionalized electrodes, faradaic systems, or complete biosensors.

The primary comparison used a single STEI-ZStat unit. The three-board comparison bounds board-to-board variation, but those boards shared a single microcontroller module and a single dummy cell, and were measured by one operator, so the result does not establish the reproducibility of independently built complete instruments. All three boards also deviated from the nominal values in the same direction and by a similar amount, so the comparison bounds board-to-board scatter but not a common-mode scale error. The 1% R_{CAL} tolerance could still produce an absolute magnitude offset larger than the cross-instrument differences reported here. Replicates were acquired in separate blocks for each instrument rather than interleaved, and the measurement environment was not controlled or recorded; temporal and environmental effects therefore cannot be separated from instrument effects. Finally, redox measurements, functionalized electrodes, and the additional electroanalytical methods implemented in firmware remain unvalidated.

VI. CONCLUSION AND FUTURE WORK

This work presented STEI-ZStat, a compact (63.1 × 45.0 mm), low-cost (US\$25.36) AD5941-based potentiostat for EIS measurements from 0.2 Hz to 200 kHz. Validated against a PalmSens4 reference and analytical models, the system demonstrated practical equivalence, with all fitted circuit parameters satisfying predefined acceptance criteria and exhibiting high inter-board reproducibility.

To address the lack of standardized evaluation in low-cost systems, we introduced a four-stage validation procedure combining reference-free screening with uncertainty-based equivalence testing. This framework provides a rigorous, standardized methodology applicable to the validation of future impedance measurement instruments.

Future work will extend validation to biological redox couples and functionalized electrodes, evaluate wider impedance ranges, and improve high-frequency phase accuracy. Additionally, we will experimentally validate the implemented voltammetric techniques and investigate Kramers–Kronig consistency analysis as an advanced data-validity check.

REFERENCES

- [1] J. S. Daniels and N. Pourmand, “Label-free impedance biosensors: Opportunities and challenges,” *Electroanalysis: An International Journal Devoted to Fundamental and Practical Aspects of Electroanalysis*, vol. 19, no. 12, pp. 1239–1257, 2007.
- [2] F. Lisdar and D. Schäfer, “The use of electrochemical impedance spectroscopy for biosensing,” *Analytical and bioanalytical chemistry*, vol. 391, no. 5, pp. 1555–1567, 2008.
- [3] H. S. Magar, R. Y. Hassan, and A. Mulchandani, “Electrochemical impedance spectroscopy (EIS): Principles, construction, and biosensing applications,” *Sensors*, vol. 21, no. 19, p. 6578, 2021.
- [4] C. M. Brett, “Electrochemical impedance spectroscopy in the characterisation and application of modified electrodes for electrochemical sensors and biosensors,” *Molecules*, vol. 27, no. 5, p. 1497, 2022.
- [5] A. C. Lazanas and M. I. Prodromidis, “Electrochemical impedance spectroscopy—a tutorial,” *ACS measurement science au*, vol. 3, no. 3, p. 162, 2023.
- [6] R. Pruna, F. Palacio, A. Baraket, N. Zine, A. Streklas, J. Bausells, A. Errachid, and M. López, “A low-cost and miniaturized potentiostat for sensing of biomolecular species such as TNF- α by electrochemical impedance spectroscopy,” *Biosensors and Bioelectronics*, vol. 100, pp. 533–540, 2018.
- [7] F. Burgos-Flórez, A. Rodríguez, E. Cervera, V. Zucolotto, M. Sanjuán, and P. J. Villalba, “TBISTAT: An open-source, wireless portable, electrochemical impedance spectroscopy capable potentiostat for the point-of-care detection of s100b in plasma samples,” *PloS one*, vol. 17, no. 2, p. e0263738, 2022.
- [8] K. A. Bautista, E. Madsen, S. D. Riegle, and J. C. Linnes, “HELPStat: a handheld, EIS-enabled, low-cost, and portable potentiostat,” *ACS Electrochemistry*, vol. 1, no. 3, pp. 386–394, 2025.
- [9] D. Bill, M. Jasper, A. Weltin, G. A. Urban, S. J. Rupitsch, and J. Kieninger, “Electrochemical methods in the cloud: FreiStat, an IoT-enabled embedded potentiostat,” *Analytical chemistry*, vol. 95, no. 35, p. 13003, 2023.
- [10] A. A. Rowe, A. J. Bonham, R. J. White, M. P. Zimmer, R. J. Yadgar, T. M. Hobza, J. W. Honea, I. Ben-Yaacov, and K. W. Plaxco, “Cheapstat: An open-source, “do-it-yourself” potentiostat for analytical and educational applications,” *PloS one*, vol. 6, no. 9, p. e23783, 2011.
- [11] I. Anshori, I. Syafalni, C. Reivan, I. F. Ramadhan, T. M. Jonathan, R. I. Sari, Uperianti, T. Adiono, C.-Y. Chang, and A. Surawijaya, “GaneStat—a comprehensive design and modular analysis of portable, low-cost, and high-accuracy potentiostat,” *Plos one*, vol. 20, no. 11, p. e0335917, 2025.
- [12] I. Vamos, R. Morrison, S. Michaletzky, and V. Kertesz, “HunStat2—a simple and low-cost potentiostat with electrochemical impedance spectroscopy capability,” *Analytical Methods*, vol. 17, no. 46, pp. 9494–9499, 2025.
- [13] D. M. Jenkins, B. E. Lee, S. Jun, J. Reyes-De-Corcuera, and E. S. McLamore, “ABE-Stat, a fully open-source and versatile wireless potentiostat project including electrochemical impedance spectroscopy,” *Journal of The Electrochemical Society*, vol. 166, no. 9, pp. B3056–B3065, 2019.
- [14] Y. Matsubara, “A small yet complete framework for a potentiostat, galvanostat, and electrochemical impedance spectrometer,” 2021.
- [15] E. W. Brown, M. W. Glasscott, K. Conley, J. Barr, J. D. Ray, L. C. Moores, and A. Netchaev, “ACEstat: a diy guide to unlocking the potential of integrated circuit potentiostats for open-source electrochemical analysis,” *Analytical chemistry*, vol. 94, no. 12, pp. 4906–4912, 2022.
- [16] A. Devices, “AD5940/AD5941: High precision, impedance, and electrochemical front end,” *Data Sheet, Rev. D*, 2024.
- [17] Analog Devices, Inc., “AD5940_ECSns_EIS example code,” https://github.com/analogdevicesinc/ad5940-examples/tree/de009cf/examples/AD5940_ECSns_EIS, 2026, commit: de009cf, Accessed: Aug. 09, 2026.
- [18] M. Kasper, A. Leike, J. Thielmann, C. Winkler, N. A.-Z. R-Smith, and F. Kienberger, “Electrochemical impedance spectroscopy error analysis and round robin on dummy cells and lithium-ion-batteries,” *Journal of Power Sources*, vol. 536, p. 231407, 2022.
- [19] M. Finšgar, D. Majer, U. Maver, and T. Maver, “Reusability of SPE and Sb-modified SPE sensors for trace Pb (II) determination,” *Sensors*, vol. 18, no. 11, p. 3976, 2018.
- [20] Gamry Instruments, “Universal dummy cell 4 operator’s manual, Rev. 1.0,” Gamry Instruments, Inc., Warminster, PA, USA, Feb. 2011, accessed: Aug. 09, 2026. [Online]. Available: <https://www.gamry.com/assets/Uploads/Universal-Dummy-Cell-Operators-Manual.pdf>.
- [21] T. I. Wong and X. Zhou, “Cyclic voltammetry signal analyses for equivalent electric circuits consisting of multiple resistors and capacitors,” *Measurement Science and Technology*, vol. 35, no. 4, p. 045018, Jan. 2024.
- [22] M. Newville, T. Stensitzki, D. B. Allen, M. Rawlik, A. Ingargiola, and A. Nelson, “LMFIT: Non-linear least-square minimization and curve-fitting for python,” *Astrophysics Source Code Library*, pp. ascl-1606, 2016.
- [23] KEMET Electronics Corp., “Commercial off-the-shelf (COTS) for higher reliability applications, X7R dielectric, 6.3–250 VDC,” Datasheet C1027_X7R_COTS_SMD, Feb. 2025.